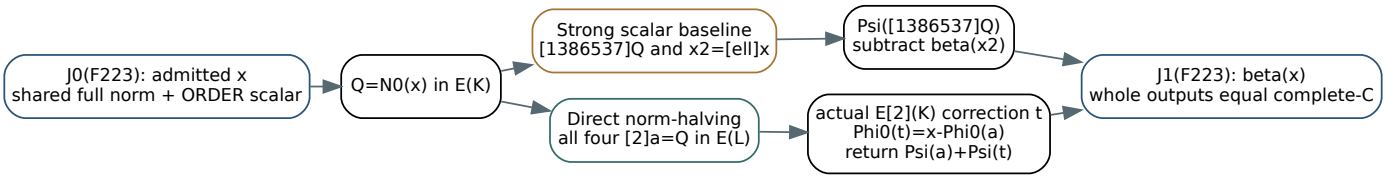
Complete norm-halving transfer

FINITE CORRECTNESS / UNSIGNED · computational advantage open

The new direct evaluator and stronger scalar baseline produce the immutable complete-C degree13 Jacobian transfer on the declared finite panel. Native05 passes all four unfiltered tests; all892 inputs stay unchanged. Entire prior three native JSON records are identical, with no ignored fields. (7ff793; earlier52c01c.)



Solid arrows describe native-tested routes at the fixed instance. They do not certify a general geometric construction or a priced cost improvement.

Correct the half before evaluation

For every admitted x, the existing odd/cofactor decomposition supplies a full elliptic lift a0 in E(L), with Phi0(a0)=x and 2a0=N0(x). Every half a of N0(x) differs from a0 by E[2](K). Look up t with Phi0(t)=x−Phi0(a), then use the explicit quotient square beta Phi0=Psi:

```
Psi(a)+Psi(t)=Psi(a+t)=beta(x).
```

Every four actual halves is checked. The runtime includes quadratic Phi/Psi, the source isogeny, dictionary, curve checks, origin subtraction and rational output check. The direct transfer core has no CRT or half-odd scalar. Common admission still includes the full-order Jacobian scalar. Full inverse still uses the Jacobian scalar3413013.

Declared checks and negative controls

448 exceptional classes;48 mixed cases;40 public sector/direction rows;96 addition pairs; both new-map compositions equal[13]; full new inverse; original independent public source recovery. The48 coordinate targets match the literal selector, with infinity, nonzero two-torsion, generic and non-K halves. Changed quartic constant, zero denominator and incorrect coordinate-F controls are retained. Both deliberately shifted origins fail original odd-membership validation. The missing whole-record canary exits101. (7ff793/d91395.)

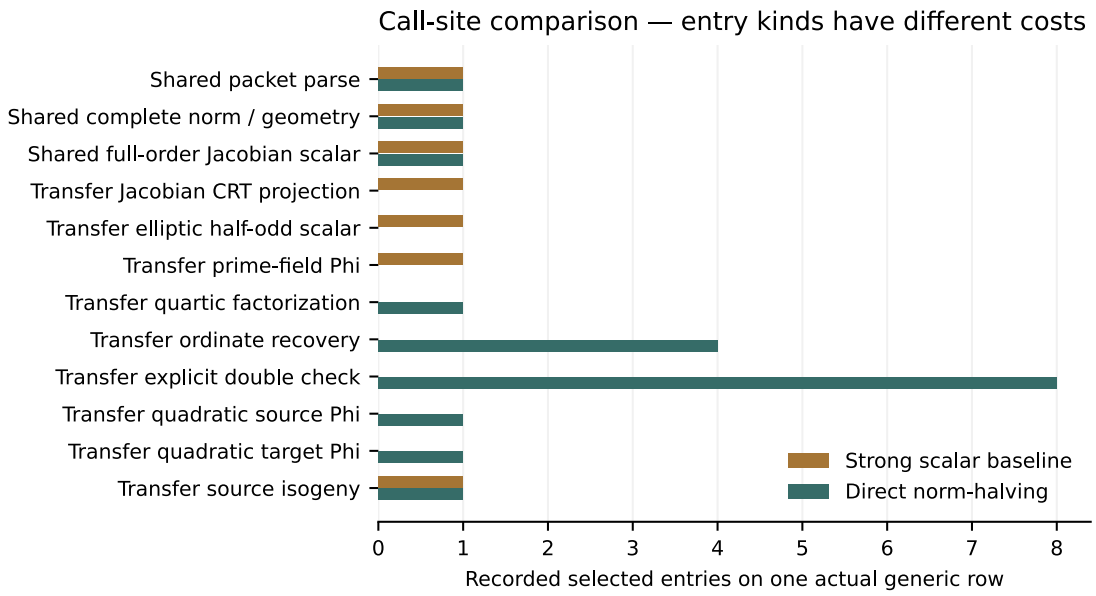
Exact fields and models

K=F223[z]/(z³−3), L=F223[w]/(w⁶−3), z↦w²; K-Frobenius on L is F³. E: a=(214,144,205), b=(137,218,9); E13: a=(196,168,69), b=(8,82,207). Full quartics and origins retain the frozen complete-C identities.

J0: Ciani-Jac-pi0-v2-3ba99ad5742ad62540f8addbbe153572696d2e595f64cd2c1b0b52e51ff351a4

J1: Ciani-Jac-odd-product-v1-7ea9d0f9d1a48a48489425beab48274fca7bd0848e47a0bac3a027c01601e2c4

Calls, proofs and remaining obligations



The chart renders actual selected entries from one recorded generic input. Rows have different primitive costs; no totals, ratios or speedup inference. Complete literal-selector, context, field, Picard, admission and verification work remains included and unpriced.

Accounting that stays in scope

Both methods include packet parsing, complete validation norm and full-order Jacobian admission. The direct setup records two elliptic two-torsion factorizations, eight prime-field Phi calls, eight source-isogeny calls and two quadratic origin traces. The strong baseline includes immutable complete-C setup and its additional isogeny context. Direct generic evaluation has four ordinate recoveries and eight explicit double checks; literal-selector internal checks remain unpriced. The448 exceptional rows take446 generic branches and two infinity branches. Nonzero two-torsion coordinates are tested separately. No timing claim is made.

Faithful conditional proofs

Both unchanged Lean4.34.1/4.35.0 gates accept ten declarations, zero trusted research axioms; named roots use standard foundations. Nine new exact statement bindings bring the partial count to82; the existing abscissa identity is reused. Proofs cover the universal ordinate identity, differences of halves, Frobenius-square fixing, odd lift, full lift from actual cofactor data, correction existence, evaluator equality, independence and the stronger scalar baseline. (3ba28f.)

The actual field/order/Weil-restriction/quotient/Picard/cofactor bridges remain written/native. The full lift is derived from odd plus actual cofactor lifts, not geometric surjectivity alone. Exact-Z expansion and the unchanged original certificate checker accept both polynomials; deleting −2B gives an actual original-checker counterexample101. The certificate’s Lean pointer does not run Lean. (d91395.)

Sources, preserved failures and next task

Sutherland2025, Lecture2 §2.2 (duplication) and Lecture5, Theorems5.8/5.25. The original project already had the halving quartic, selector and ring proof; no novelty claim. Local literal source, hashes, full derivation and proof map accompany this report.

Build01, constructor03, Lean01/02 and all earlier negative records remain retained. Native04/05 whole sources, outputs and actual binaries have complete indices. Complete primitive coverage, admitted matched algorithm references, full geometric bridges and a genuinely nondual degree13 continuation stay open. Live measurement admission, root regression, signing/merge requirements and locked KB/vault limits persist.